

Notes: Huang, Li, Ma, and Xu (AER, 2017)

Hayek, Local Information, and Commanding Heights: Decentralizing State-Owned Enterprises in China

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1 Research question and core contribution

1.1 What the paper asks

The paper studies why a higher-level Chinese government (central/provincial/municipal) sometimes **delegates (decentralizes) control rights** over a state-owned enterprise (SOE) to a lower-level government. The key empirical object is a decentralization event: an SOE’s oversight affiliation moves *down* the administrative hierarchy.

1.2 Why this matters (economic logic)

The paper is an empirical test of Hayek’s (1945) classic insight: if the economic problem is adaptation to time-and-space-specific conditions, then control should be delegated to those closer to the facts. The Chinese SOE setting is useful because: (i) the government hierarchy is explicit, (ii) oversight transfers are observable, and (iii) the state remains an important owner while pursuing both efficiency and strategic goals.

1.3 Main conclusion (preview)

The evidence supports a **Hayek (local information) mechanism**: SOEs that are farther from their oversight government are more likely to be decentralized, and this *distance–decentralization link* becomes stronger when communication costs are higher and when firm performance is more heterogeneous (local information is more valuable). However, the link is **muted for central SOEs in strategic industries**, consistent with a “commanding heights” motive. Alternative agency/capture stories receive little support.

2 Institutional setting and definitions

2.1 Government hierarchy and “decentralization”

China’s government hierarchy is: central → provincial → municipal → county. An SOE is “affiliated” with (overseen by) one level. A **decentralization** occurs when the SOE’s affiliation level moves down (e.g., central to province, province to city).

2.2 Two distance concepts

The paper uses physical distance as a proxy for information frictions.

- **Distance to oversight government** (DisOS): distance from the SOE to the seat of the government that currently oversees it.
- **Distance to lower government** (DisLower): distance from the SOE to the potential recipient lower-level government (relevant for local-capture lobbying stories and also for the efficiency gain from delegating to a nearby lower government).

3 Conceptual framework: competing hypotheses and their testable implications

The paper contrasts four families of explanations. Write the probability of decentralization as increasing in the net benefit of delegating control.

3.1 (H1) Hayek (local information) conjecture

Mechanism: A more distant oversight government has weaker access to firm-specific local information. Delegating to a closer government improves adaptation and efficiency.

Predictions:

1. $\text{DisOS} \uparrow \Rightarrow \Pr(\text{Decen} = 1) \uparrow$ (positive distance–decentralization link).
2. The link is **stronger** when local information is more valuable:
 - *Higher firm-performance heterogeneity* (e.g., higher dispersion of TFP/ROS in an industry-year).
 - *Higher communication costs* (e.g., lower road density).

3.2 (H2) Commanding-heights conjecture

Mechanism: For strategic industries, the central government prioritizes strategic/nonprofit objectives and control over purely efficiency gains. Delegation becomes less desirable.

Prediction:

$\text{Strategic industry} \times \text{Central oversight} \Rightarrow \text{weaker (less positive) DisOS effect.}$

3.3 (H3) Oversight-agency (rent extraction by the oversight government)

Mechanism: Nearby SOEs are easier to monitor and extract rents from; distance reduces private benefits of control. If rents are high, the oversight government has more incentive to keep nearby SOEs.

Unique prediction:

$\text{DisOS} \times (\text{rents/corruption}) \text{ should be positive.}$

(High rents make proximity relatively more attractive to the incumbent overseer, strengthening the distance gradient.)

3.4 (H4) Local capture (lobbying/collusion by lower governments)

Mechanism: Lower governments may lobby for decentralization to collude with managers. Collusion is easier when the SOE is close to the lower government.

Unique prediction:

$\text{DisLower} \times (\text{rents/corruption})$ should be negative.

(High rents strengthen the role of proximity in collusion, making decentralization especially likely when DisLower is small.)

Table 2 in the paper summarizes the predicted signs and indicates which predictions match the empirical evidence.

4 Data and measurement

4.1 Dataset and sample construction

The paper uses the Annual Survey of Industrial Firms (ASIF), 1998–2007. Baseline regressions focus on SOEs (state ownership share $> 30\%$) and exclude post-decentralization observations (to avoid mixing pre- and post-treatment periods in the selection equation). The baseline sample excludes county-or-below SOEs (cannot be decentralized) and requires multiple consecutive years of data.

4.2 Key variables (from Table 1)

- Decen_{ijkt} : indicator that firm i in industry j , overseen by government k , in year t is decentralized (event-time definition).
- Distance_{ik} : log distance (km) between firm i and the seat of its oversight government k (often lagged in regressions).
- **Communication cost proxies**: province road mileage/density (lower road density $\Rightarrow$ higher communication costs).
- **Firm-performance heterogeneity**: dispersion (standard deviation) of TFP or ROS in a 3-digit industry-year. TFP is computed using multiple methods (OLS residual, Olley–Pakes, and index-number).
- **Strategic industries**: two classifications (Strategic I, Strategic II) capturing defense/energy/utilities and related sectors.
- **Rents/corruption proxies**: corruption cases per capita, entertainment/travel expenditures share, firm profitability (ROS), firm average wage, industry HHI.
- **Instrument**: TFC_i indicates firms established during China’s Third Front Construction in TFC regions (historical relocation of SOEs inland), used to instrument distance.

5 Empirical strategy: all models in the paper

5.1 Model 1: baseline probit (Equation (1))

The main specification is a probit model for decentralization:

$$\Pr(\text{Decen}_{ijkt} = 1) = \Pr\left(\delta \text{Distance}_{ik} + X'_{ijk,t-1}\beta + Z'_{t-1}\alpha + \theta_j + \rho_k + \gamma_t + \varepsilon_{ijkt} > 0\right), \quad (1)$$

where X includes lagged firm covariates (size, ROS, firm importance, full state ownership), Z includes lagged province covariates (GDP per capita, unemployment, state-sector share), and $\theta_j, \rho_k, \gamma_t$ are industry, oversight-government, and year fixed effects.

Reporting convention. In the tables, the paper reports **marginal effects** from probit and clusters standard errors at the oversight-government level.

5.2 Model 2: event-history / hazard model (robustness)

Because decentralization is an absorbing event (once decentralized, it stays so), the paper also estimates a duration model. In Table 3, column (5) reports a **Cox proportional hazard model**:

$$h_i(t \mid W_{i,t-1}) = h_0(t) \exp(W'_{i,t-1}\kappa), \quad W_{i,t-1} \equiv [\text{Distance}_{i,\cdot}, X_{i,t-1}, Z_{t-1}, \text{FE}]. \quad (1)$$

A positive coefficient on distance implies a higher instantaneous hazard of decentralization.

5.3 Model 3: multinomial logit with restructuring as an outside option (robustness)

To ensure that distance is not simply proxying for broader restructuring policies, the paper models three choices in a year:

$$\{\text{no change (base), decentralize, restructure}\}.$$

Let option $m \in \{0, 1, 2\}$ have latent utility $U_{im} = V_{im} + \eta_{im}$ with i.i.d. type-I extreme value shocks. Then

$$\Pr(m \mid V) = \frac{\exp(V_{im})}{\sum_{\ell} \exp(V_{i\ell})}. \quad (2)$$

The table reports marginal effects on the probability of choosing “decentralize” vs “restructure.”

5.4 Model 4: placebo test (distance that should not matter)

To rule out spurious geography correlations, the paper constructs a *placebo distance* (distance to an economically large city that is not the overseer) and re-estimates the decentralization regression. A null placebo result supports the interpretation that what matters is the oversight-information distance.

5.5 Model 5: instrumental variables (2SLS) using Third Front Construction

Distance could be endogenous if poorly performing/unimportant SOEs are systematically located farther away. The paper uses a historical relocation program as an instrument.

First stage.

$$\text{Distance}_{i,t-1} = \pi_0 + \pi_1 \text{TFC}_i + X'_{i,t-1} \pi_2 + Z'_{t-1} \pi_3 + \text{FE} + u_{i,t}. \quad (3)$$

Second stage (linear probability / 2SLS).

$$\text{Decen}_{i,t} = \beta_0 + \beta_1 \widehat{\text{Distance}}_{i,t-1} + X'_{i,t-1} \beta_2 + Z'_{t-1} \beta_3 + \text{FE} + \varepsilon_{i,t}. \quad (4)$$

The paper reports weak-instrument-robust inference (Anderson–Rubin) and the Kleibergen–Paap F statistic.

5.6 Model 6: interaction tests for mechanisms and alternative hypotheses

To test the Hayek and commanding-heights conjectures, the paper augments (1) with interactions, e.g.

$$\Pr(\text{Decen} = 1) = \Phi\left(\delta \text{Distance} + \eta (\text{Distance} \times M) + \lambda M + \dots\right), \quad (5)$$

where M is a mechanism proxy (communication costs, heterogeneity, strategic-industry status, corruption/rents). To test local-capture stories, it also includes DisLower and interactions with rent proxies.

6 Interpreting every table in the paper

6.1 Table 1: variable descriptions (measurement map)

Table 1 is the **dictionary** of the empirical design. The important conceptual mapping is:

Information frictions $\leftrightarrow$ Distance, road density,
Value of local information $\leftrightarrow$ TFP/ROS dispersion,
Strategic control $\leftrightarrow$ strategic industry indicators,
Agency/capture rents $\leftrightarrow$ corruption and rent proxies.

As a reading strategy: keep Table 1 open while reading Tables 3–5, because the paper’s identification is “distance + interactions.”

6.2 Table 2: sign predictions and which mechanisms survive

Table 2 summarizes the empirical discriminators among hypotheses. The key takeaway is that the evidence matches:

- **Hayek:** DisOS positive; DisOS $\times$ heterogeneity positive; DisOS $\times$ communication-cost proxy (road density) negative.
 - **Commanding heights:** strategic industries $\times$ central oversight dampen the distance effect.
- and fails to match the **unique** predictions of the agency/capture alternatives (rent interactions).

TABLE 1—VARIABLE DESCRIPTIONS

Firm-level and industry-level variables	
Decentralized	Dummy variable which equals to one if a firm's affiliation-level is changed to a lower level of the government, zero otherwise. <i>Source:</i> Annual Survey of Industrial Firms (ASIF).
Different city	Dummy variable which equals to one if being located in a different city from the seat of the government with which it is affiliated, and zero otherwise.
Distance	The logarithm of the physical distance (in kilometers) between the firm and the seat of the oversight government. <i>Sources:</i> ASIF and Geographic Information System.
Placebo distance	Distance from Shanghai for central SOEs; distance from the largest non-capital city within the province for provincial SOEs, and distance from the largest non-municipal-seat county-level city within the same municipality for municipal SOEs. <i>Sources:</i> ASIF and GIS.
Restructured	Dummy variable equals to one if the state share falls below 30 percent or exit from the database.
Firm asset	Log of firm asset.
ROS	Ratio of before-tax profit on sales, winsorized at tail 0.5 percents.
TFP	Total factor productivity (see online Appendix K for details). Three methods are used, including OLS, Olley-Pakes, and the index number.
Firm importance	Ratio of the SOE's value added to the total value added of the SOEs under the same oversight government.
Fully state-owned	Dummy variable which equals to one if the state share in the firm's equity equals 100 percent.
Average wage	Log of the average wage (in thousand yuan) within the firm.
Industry-level HHI	Herfindahl index at the three-digit industry level.
TFP dispersion	Standard deviation of the firm's TFP in the three-digit-industry-year level.
ROS dispersion	Standard deviation of the firm's ROS in the three-digit-industry-year level.
TFC	Dummy variable equals to one if the firm was established during 1964–1966 or 1969–1971 in the <i>Third Front Construction Area</i> (see online Appendix I).
Fiscal, economic, and institutional variables at province, municipal, and county levels	
Provincial (municipal, county) GDP per capita	Log of annual per capita GDP in the firm's province (municipality, or county), in constant price of year 2000. <i>Source:</i> China Statistical Yearbooks (CSYs) (1998–2007).
Provincial state sector share	Share of SOE employment in total urban employment in the firm's province. <i>Source:</i> CSYs.
Provincial unemp. rate	The annual urban unemployment rate in the firm's province. <i>Source:</i> CSYs.
Provincial (municipal, county) fiscal revenue per capita	Log of the annual fiscal revenue per capita in the firm's province (municipality and county), in constant price of year 2000. <i>Sources:</i> CSYs and Financial and Statistical Materials of Municipalities, Cities, and Counties (FSMPCCs) (1998–2007).
Provincial (municipal, county) fiscal autonomy	The ratio of fiscal revenue to fiscal expenditure in the firm's province (municipality and county). <i>Source:</i> CSYs and FSMPCCs.
Road mileage	Road distance (in kilometers) per thousand people at the province level. Different levels of roads and railway mileage are translated into the equivalent of second-level road mileage according to transport capacity. <i>Sources:</i> CSYs and FSMPCCs.
Entertainment and travel cost	The average ratio of the firm's entertainment and travel cost to sales in a province. A higher value means more corruption. <i>Source:</i> World Bank Enterprises Survey for China, 2005.
Corruption cases	Graft cases filed per thousand people. <i>Source:</i> Procuratorial Yearbooks of China.

TABLE 2—SUMMARY OF PREDICTIONS FROM KEY HYPOTHESES

Predictions from the hypotheses	Hayek information	Commanding heights	Oversight agency	Local capture
Distance to the oversight government (DisOS _{gov})	Positive ^{YES}		Positive ^{YES}	
DisOS _{gov} $\times$ firm heterogeneity	Positive ^{YES}			
DisOS _{gov} $\times$ communication costs	Positive ^{YES}		Positive ^{YES}	
DisOS _{gov} $\times$ strategic industries $\times$ central government		Negative ^{YES}		
DisOS _{gov} $\times$ rents			Positive ^{NO}	
Distance to lower government (DisL _{lower})	Negative ^{YES}			Negative ^{YES}
DisL _{lower} $\times$ rents				Negative ^{NO}

Notes: YES: results consistent with predictions. NO: results inconsistent with predictions. Empty cell: no prediction.

6.3 Table 3: determinants of decentralization (baseline and robustness)

Panel A (baseline). Columns (1)–(4) (probit marginal effects) show Distance_{*t*−1} is positive and highly significant across: whole sample, central SOEs, provincial SOEs, and municipal SOEs. Thus, **farther SOEs are more likely to be decentralized**, consistent with the local-information mechanism.

Controls have intuitive signs: smaller, less profitable, less important, and fully state-owned SOEs are more likely to be decentralized, consistent with decentralization being used as a reform tool for weaker firms.

Column (5) (hazard). The Cox hazard model yields a positive distance coefficient, indicating that distance raises the *hazard* of decentralization. This is consistent with treating decentralization

as an absorbing event and supports robustness to modeling choices.

Columns (6)–(7) (multinomial logit). When restructuring is included as an alternative option: distance remains strongly related to decentralization, while the distance effect for “restructured” is small/insignificant. This matters: **distance predicts decentralization specifically**, not generic restructuring.

Panel B (robustness). Additional controls and alternative distance definitions do not overturn the positive distance effect. Subsample evidence (fully state-owned vs partially privatized) still shows a positive distance gradient.

6.4 Table 4: placebo and IV evidence (addressing alternative interpretations)

Placebo test (cols. (1)–(2)). The placebo distance has coefficients near zero and statistically insignificant. Interpretation: the baseline result is not driven by generic “being far from big cities.”

IV (cols. (3)–(4)). First stage: TFC significantly increases distance. Second stage: instrumented distance remains positive and significant for decentralization. This supports a causal reading that exogenous increases in distance raise the likelihood of decentralization, consistent with the paper’s informational interpretation.

TABLE 3—DETERMINANTS OF DECENTRALIZATION

	Probit				Hansen		Multinomial logit	
	Whole sample	Control SOEs	Provincial SOEs	Municipal SOEs	Whole sample	Control SOEs	Decentralized _{it}	Restructured _{it}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Panel A. Baseline results</i>								
Distance _{it}	0.0024 (0.0007)	0.0047 (0.0017)	0.0037 (0.0008)	0.0044 (0.0005)	0.2099 (0.0454)	0.0053 (0.0008)	0.0011 (0.0010)	
Log(firm asset) _{it}	-0.0029 (0.0002)	-0.0049 (0.0003)	-0.0030 (0.0003)	-0.0037 (0.0003)	-0.1454 (0.0320)	-0.0025 (0.0002)	-0.0109 (0.0011)	
ROS _{it}	-0.0002 (0.0015)	-0.0171 (0.0065)	-0.0129 (0.0029)	-0.0044 (0.0017)	-0.5428 (0.0017)	-0.0074 (0.0015)	-0.0683 (0.0065)	
Firm importance _{it}	-0.0066 (0.0205)	-0.2243 (0.3033)	-0.1461 (0.1449)	-0.0248 (0.0152)	-2.5841 (1.1152)	-0.0199 (0.0187)	-0.0716 (0.0200)	
Fully state-owned _{it}	-0.0069 (0.0013)	-0.0129 (0.0051)	-0.0147 (0.0023)	0.0001 (0.0013)	-0.3387 (0.0723)	-0.0059 (0.0013)	-0.0204 (0.0065)	
GDP per capita _{it}	0.0061 (0.0004)	-0.0026 (0.0003)	-0.0004 (0.0033)	0.0020 (0.0234)	0.3009 (0.2431)	0.0043 (0.0043)	0.0007 (0.0154)	
State sector share _{it}	-0.0042 (0.0009)	-0.0019 (0.0219)	0.0767 (0.0625)	0.0020 (0.0470)	-0.2776 (1.6384)	-0.0140 (0.0270)	0.2386 (0.1969)	
Unemployment rate _{it}	-0.1128 (0.1108)	-0.2042 (0.2195)	-0.1744 (0.4464)	-0.1104 (0.2865)	-7.2025 (7.1847)	-0.0258 (0.1222)	-0.8419 (0.4271)	
Government, year, and industry dummies	YES	YES	YES	YES	YES	YES	YES	
Observations	69,785	11,171	20,356	38,258	69,785	69,785	69,785	
Pseudo R ²	0.115	0.085	0.111	0.119	0.077			
<i>Panel B. Robustness checks: probit</i>								
Distance _{it}	0.0045 (0.0017)	0.0025 (0.0010)	0.0047 (0.0006)	0.0063 (0.0008)	0.0066 (0.0014)			
Controls	YES	YES	YES	YES	YES			
Additional controls	YES	YES	YES	YES	YES			
Observations	11,171	19,362	32,074	54,161	15,024			
Pseudo R ²	0.086	0.116	0.104	0.113	0.143			

TABLE 4—DETERMINANTS OF DECENTRALIZATION: PLACEBO TEST AND THIRD FRONT CONSTRUCTION AS IV

	Placebo test		2SLS	
	Whole sample	Excluding small placebo city	First stage Distance	Second stage Decentralized _{it}
	(1)	(2)	(3)	(4)
Placebo distance _{it}	0.0005 (0.0009)	0.0002 (0.0012)		
Distance _{it}			0.0240 (0.0124)	
TFC			0.2416 (0.1018)	
Controls	YES	YES	YES	YES
Observations	69,785	36,742	69,785	69,785
Kleibergen-Paap F-statistic			5.627	
Anderson-Rubin F-statistic			3.83	
(p-value)			[0.0504]	
90 percent confidence interval			[0.0050, 0.0537]	

Notes: Standard errors clustered at the oversight-government level are reported in the parentheses. Columns 1–2 report results on placebo distance using the probit model. Placebo distance is defined as the distance from Shanghai for the control SOEs, the distance from the largest city within the province (except for the provincial capital) for provincial SOEs, and the distance from the largest county-level city within the municipality (except for the municipal seat) for municipal SOEs. Column 2 excludes the provincial and municipal SOEs located in the same jurisdiction with the jurisdictional capital as the sole agglomeration city. Columns 3–4 report results using TFC as the instrument. Column 3 reports the first stage results. Column 4 reports the linear probability IV estimates (i.e., 2SLS) and the associated test statistics. The control variables are the same as in Table 3.

TABLE 5—DETERMINANTS OF DECENTRALIZATION: TESTING ALTERNATIVE HYPOTHESES

	Province road mileage _{it}	Province road mileage _{it} × on	ROS dispersion _{it}	TFC OLS dispersion _{it}	TFC OLS dispersion _{it} × on	TFC Index dispersion _{it}
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Panel A. Key indicators of communication costs or firm performance heterogeneity</i>						
Distance _{it}	0.0006 (0.0010)	0.007 (0.0011)	0.002 (0.0009)	0.0016 (0.0019)	-0.0021 (0.0020)	0.0009 (0.0018)
Distance _{it} × Key indicators	-0.0021 (0.0004)	-0.0017 (0.0006)	0.0117 (0.0045)	0.0026 (0.0011)	0.0048 (0.0012)	0.0012 (0.0011)
Key indicators	0.0132 (0.0033)	0.0113 (0.0040)	-0.0054 (0.0277)	-0.0116 (0.0099)	-0.0132 (0.0066)	-0.0111 (0.0081)
Controls	YES	YES	YES	YES	YES	YES
Observations	69,785	69,785	69,785	69,784	61,660	69,784
Pseudo R ²	0.118	0.116	0.116	0.116	0.114	0.116
<i>Panel B. Interactions of distance with strategic industries</i>						
Distance _{it}	0.0055 (0.0007)	0.0048 (0.0017)	0.0046 (0.0005)	0.0056 (0.0006)	0.0056 (0.0017)	0.0046 (0.0005)
Strategic industries I × Distance _{it}	-0.0011 (0.0007)	-0.0047 (0.0016)	0.0008 (0.0011)			
Strategic industries II × Distance _{it}				-0.0011 (0.0006)	-0.0052 (0.0018)	0.0002 (0.0007)
Controls	YES	YES	YES	YES	YES	YES
Observations	69,785	11,171	38,614	69,785	11,171	38,614
Pseudo R ²	0.116	0.088	0.115	0.113	0.087	0.114
<i>Key indicators of corruption heterogeneity</i>						
Entertainment and travel cost	(1)	(2)	(3)	(4)	(5)	(6)
Corruption cases _{it}						
Distance to oversight gov _{it}	0.0055 (0.0007)	0.0082 (0.0022)	0.0048 (0.0016)	0.0054 (0.0006)	0.0060 (0.0011)	0.0056 (0.0006)
Distance to oversight gov _{it} × Key indicators	-0.2011 (0.1520)	0.0140 (0.0465)	-0.0074 (0.0033)	-0.0052 (0.0005)	-0.0052 (0.0005)	-0.0052 (0.0005)
Distance to Lower gov _{it} × Key indicators	-0.0010 (0.0004)	0.0006 (0.0012)	0.0009 (0.0022)	-0.0010 (0.0003)	-0.0011 (0.0010)	-0.0011 (0.0006)
Distance to Lower gov _{it} × Key indicators	-0.1177 (0.0753)	-0.0008 (0.0020)	-0.0008 (0.0000)	-0.0001 (0.0000)	0.0024 (0.0004)	0.0024 (0.0004)
Key indicators	2.1216 (0.0666)	-0.0647 (0.3534)	0.0002 (0.0011)	-0.0012 (0.0021)	0.0086 (0.0011)	0.0086 (0.0011)
Controls	YES	YES	YES	YES	YES	YES
Observations	69,785	69,785	66,173	69,785	69,157	69,785
Pseudo R ²	0.116	0.116	0.117	0.116	0.117	0.116

Notes: Standard errors clustered at the oversight-government level are reported in the parentheses. This table reports the marginal effect from probit regressions. The control variables are the same as in Table 3. In panel B, Strategic industries I includes (i) oil and gas, petroleum, (ii) nuclear fuel, aviation and aerospace, arms and ammunition, (iii) electricity, heat, gas, and water supply. Strategic industries II includes Strategic industries I, plus (iv) automobile, locomotive, and ship. See online Appendix I for details about the construction of strategic industries.

6.5 Table 5: tests of mechanisms and alternative hypotheses

Panel A: communication costs and performance heterogeneity

Communication costs (cols. (1)–(2)). Distance × road mileage is negative: when road infrastructure is better (communication cheaper), the marginal effect of distance is smaller. Equivalently,

distance matters more when communication costs are higher. This is consistent with Hayek and also (in principle) with oversight-agency stories.

Performance heterogeneity (cols. (3)–(6)). Distance $\times$ dispersion of ROS/TFP is positive and significant across multiple dispersion measures. This is the **unique** Hayek prediction: local information is more valuable when performance is heterogeneous. Thus, this panel is the strongest discriminator in favor of the Hayek mechanism.

Panel B: commanding heights

Strategic-industry interactions are negative, especially for **central SOEs**. Interpretation: when strategic objectives are salient, the central government is less willing to delegate, so the informational force of distance is muted. This matches the “commanding heights” conjecture.

Panel C: rents/corruption and agency/capture explanations

Panel C includes both DisOS and DisLower and their interactions with rent/corruption proxies. While DisOS is positive and DisLower is negative (both can be consistent with Hayek), the **interaction terms with rents are mostly insignificant**. Therefore, the evidence does *not* support the distinctive rent-interaction predictions of:

- oversight-agency (positive DisOS $\times$ rents), or
- local capture (negative DisLower $\times$ rents).

Overall, the rent-based agency/capture channels appear second-order in this setting.

7 Short summary

1. **Distance predicts delegation:** SOEs that are farther from their oversight government are more likely to be decentralized.
2. **Mechanism evidence supports Hayek:** the distance effect strengthens with higher communication costs and higher performance heterogeneity.
3. **Strategic control matters:** central SOEs in strategic industries exhibit a muted distance effect (“commanding heights”).
4. **Agency/capture stories are not first-order:** rent-related interaction predictions are mostly not supported.

8 Conclusion

8.1 Summary

This paper uses the decentralization of Chinese state-owned enterprises (SOEs) to test Hayek’s core idea that *local information* is central to the efficient organization of production. The main

conclusion is that **information frictions systematically predict decentralization**: SOEs that are **farther** from their oversight government are **more likely** to be decentralized, and this distance effect is **stronger when communication is harder** and when **firm performance is more heterogeneous** (when local information is more valuable). At the same time, decentralization is **constrained by “commanding heights” considerations**: for **central SOEs in strategic industries**, the distance effect is **muted or nearly shut down**. Finally, the paper evaluates alternative agency-cost explanations (e.g., rent extraction, local capture) and finds **little support** for them in the data.

8.2 Key empirical conclusions (with quantitative anchors)

1. **Baseline distance effect: farther SOEs decentralize more.** In the baseline probit (Table 3, Panel A, col. (1)), the marginal effect of log distance to the oversight government is

$$\hat{\beta}_{Dist} \approx 0.0054 \quad (s.e. \ 0.0007),$$

implying a sizable increase in decentralization probability as distance rises. Using the paper’s summary, a one-standard-deviation increase in distance raises decentralization probability by about **1.3 percentage points** (about **9%** of the standard deviation of the decentralization indicator). The result holds in subsamples by oversight level (central/provincial/municipal) and under alternative specifications (Cox hazard; multinomial logit distinguishing decentralization vs restructuring).

2. **Not an agglomeration artifact (placebo test).** A placebo “distance to an irrelevant city” has essentially zero effect (Table 4, cols. (1)–(2)):

$$\hat{\beta}_{Placebo} \approx 0.0005 \ (0.0009) \quad \text{or} \quad 0.0002 \ (0.0012),$$

supporting the interpretation that the main distance effect is not simply capturing “being remote from major cities.”

3. **Addressing endogeneity with an IV: Third Front Construction (TFC).** Using TFC as an instrument for distance (Table 4, cols. (3)–(4)):

$$\text{First stage: } \hat{\pi} \approx 0.2416 \ (0.1018), \quad \text{Second stage: } \hat{\beta}_{Dist}^{IV} \approx 0.0240 \ (0.0124).$$

The weak-instrument-robust Anderson–Rubin test rejects at about the 5% level (reported $p \approx 0.050$), and the 90% confidence interval is approximately $[0.0050, 0.0537]$.

4. **Hayek mechanism evidence: communication costs amplify the distance effect.** Let the marginal effect of distance depend on a proxy for communication costs C :

$$\frac{\partial \Pr(Decentralize)}{\partial Distance} = \beta_{Dist} + \beta_{Dist \times C} C.$$

Using road mileage (a transportation/communication proxy), Table 5 Panel A shows

$$\hat{\beta}_{Dist \times Road} < 0 \quad (\text{e.g., } -0.0021 \text{ (0.0004)}),$$

implying that **better infrastructure (lower communication costs) attenuates** the role of distance. The paper summarizes that increasing road density by one standard deviation reduces the marginal effect of distance by about **28%**.

5. **Hayek unique prediction: performance heterogeneity amplifies the distance effect.** Table 5 Panel A also shows that the distance effect is **stronger when industry performance dispersion is higher**, e.g. with ROS dispersion:

$$\hat{\beta}_{Dist \times Disp} \approx 0.0117 \text{ (0.0045)} > 0,$$

and similarly with multiple TFP-dispersion measures. The paper summarizes that a one-standard-deviation increase in Olley–Pakes TFP dispersion raises the marginal effect of distance by about **23%**. This is a distinctive implication of the local-information (Hayek) logic: when firms are more diverse, local adaptation matters more.

6. **Commanding-heights constraint: strategic central SOEs are different.** In Table 5 Panel B, the interaction between strategic-industry status and distance is **negative and significant for central SOEs**, e.g.

$$\hat{\beta}_{Strategic \times Dist} \approx -0.0047 \text{ (0.0016)} \quad \text{or} \quad -0.0052 \text{ (0.0018)}.$$

Since the baseline distance effect for central SOEs is about 0.0048–0.0056, the *net* distance effect for strategic central SOEs is near zero:

$$0.0048 - 0.0047 \approx 0.0001,$$

consistent with the idea that the central state prioritizes control in strategic sectors. The paper notes these “commanding-heights” SOEs represent about **5%** of industrial output.

7. **Agency-cost alternatives receive limited support.** Table 5 Panel C tests interactions of distance with proxies for corruption and rents. Most interaction terms are statistically insignificant; the paper emphasizes that these patterns do not fit the key sign predictions of oversight-agency or local-capture stories (and one significant interaction has the *opposite* sign).

8.3 Economic intuition: why the patterns line up

The empirical regularities align with a simple information-processing logic:

- **Distance is a proxy for information asymmetry.** When an SOE is far from its oversight government, the center has weaker access to local conditions (technology, demand, labor, suppliers), making centralized control less efficient.

- **Communication costs govern whether the center can “manage at a distance.”** If roads/transportation are poor (high communication costs), it is harder to transmit and verify information, so delegating authority becomes more attractive.
- **Heterogeneity increases the value of local knowledge.** When firms (within an industry) differ widely in productivity/performance, uniform centralized rules are more costly; decentralization allows decisions to adapt to the firm’s particular circumstances.
- **Strategic sectors generate a control–efficiency tradeoff.** Even if decentralization is efficiency-improving, the central government may retain control in “commanding heights” industries (energy, defense-related sectors, utilities, etc.), which mechanically weakens the distance mechanism there.

8.4 Takeaway

Chinese SOE decentralization follows an information logic: distance-induced information frictions increase decentralization, especially when communication is costly and performance is heterogeneous, but strategic “commanding heights” considerations shut down this mechanism for central SOEs in key industries.